GREENPATH AI

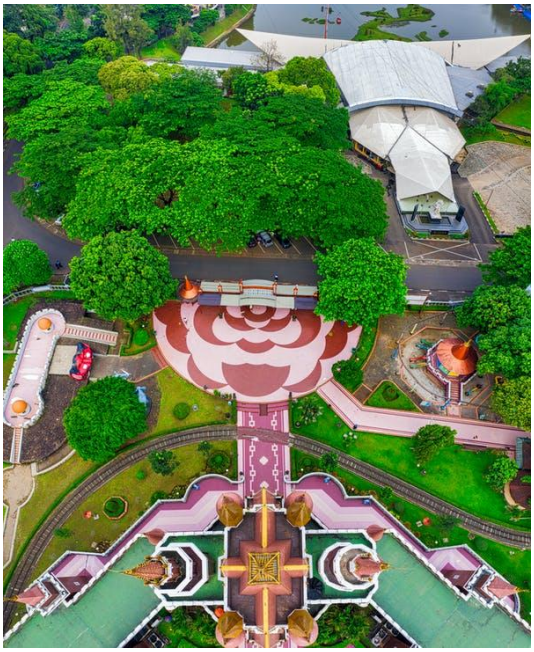
Environmental Intelligence Report

Generated: 2026-10-17 06:46 UTC
Dataset: ov_12_jpegrfe4fbca002dc496fd246ce544a8492ab2_1792219486_de9a10.jpg

EXECUTIVE SUMMARY

Trees Detected:	26
Green Cover %:	46.28%
Estimated Annual CO2 Absorption:	572 kg/year
Heat Mitigation:	3.00°C
Eco-Index Score:	87/100

1. SOURCE SATELLITE IMAGE



Source image used for environmental analysis

2. YOLOv8 TREE DETECTION



YOLOv8 tree detection and spatial bounding-box overlay

Total Trees Detected: 26

3. TREE SPECIES INTELLIGENCE

Common Name	African Tulip
Scientific Name	Spathodea campanulata
Family	Bignoniaceae
Native Region	Tropical West Africa
Habitat / Tree Type	Broadleaf Evergreen / Semi-Deciduous Canopy
Average Height	15 – 25 meters
Lifespan	50 – 80 years
Conservation Status	Least Concern (Common / Introduced)

Species Overview: A large upright tropical tree noted for its brilliant orange-scarlet bell-shaped flowers and dark green pinnate foliage.

Ecological Role: Provides vital nectar source for tropical sunbirds, bees, and arboreal pollinators.

Common Uses: Urban shade tree, ornamental street landscaping, and traditional herbal remedies for skin conditions.

Habitat & Growing Conditions:

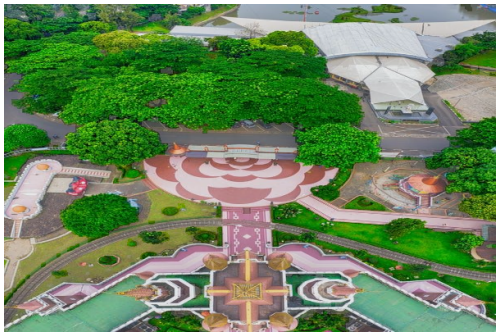
- Sunlight: Full Sun (minimum 6–8 hours direct sunlight daily).
- Water: Standard Maintenance — 25 liters weekly; natural rainfall sufficient during wet season.
- Soil: Well-drained loamy to clay-loam soil with organic leaf mulch.
- Climate / Care: N/A

4. GREEN COVER ANALYSIS

Green Cover Vegetation Percentage:	46.28%
Vegetation Cover %:	46.28%
Pavements & Built %:	53.72%



Vegetation Mask



Non-Vegetation Mask

5. RECOMMENDED PLANTING LOCATIONS

Existing Trees:	26
Estimated Additional Trees:	15
Recommended Planting Area:	2.25 hectares



AI-generated planting recommendations based on observed tree density.

● Existing Trees

● High Priority — Recommended Planting

● Medium Priority — Recommended Planting

Recommended Locations:

- Location 1: Top Right (Priority: HIGH)
- Location 2: Top Right (Priority: HIGH)
- Location 3: Top Right (Priority: HIGH)
- Location 4: Top Right (Priority: HIGH)
- Location 5: Center (Priority: HIGH)
- Location 6: Center (Priority: HIGH)
- Location 7: Center (Priority: HIGH)
- Location 8: Center (Priority: HIGH)
- Location 9: Bottom Left (Priority: HIGH)
- Location 10: Bottom Left (Priority: HIGH)
- Location 11: Bottom Left (Priority: HIGH)
- Location 12: Bottom Left (Priority: HIGH)

...and 3 more.

6. PLANTATION IMPACT & MONITORING

Trees Planted	5
Tree Survival	4
Current Green Cover	0%
CO2 Absorption Change	+88.0 kg/year
Total Expected Trees	30

Monitoring Timeline:

- **Plantation Day** (Surviving: 4, Health: 0%, Green Cover: 0%)

ENVIRONMENTAL ASSESSMENT SUMMARY

The current environmental assessment of ov_12_jpegrfe4fbca002dc496fd246ce544a8492ab2_1792219486_de9a10.jpg reveals a tree population of 26 trees. The green cover currently stands at 46.28%. The primary detected species is African Tulip. The total carbon absorption is estimated at 572 kg/year with an Eco-Index score of 87/100. There are opportunities to plant an estimated 15 additional trees in the recommended zones. Plantation monitoring currently tracks 4 surviving trees from recent afforestation efforts.